certain varieties, which re-proved certain cases of the theorems in the work of Duke, Rudnick and Sarnak [76] in a simpler manner. However, as discussed in Sect. 1.1, the most difficult—and sometimes most interesting—problem is to understand the orbit of a given point rather than the orbit of almost every point. Indeed, the solution of Oppenheim's conjecture in Sect. 1.4 by Margulis involved understanding the $\mathrm{SO}(2,1)$-orbit of a point in $\mathrm{SL}_3(\mathbb{Z})\backslash\mathrm{SL}_3(\mathbb{R})$ corresponding to the given quadratic form.

We need one more definition before we can state a general theorem in this direction. A subgroup $U < \mathrm{SL}_n(\mathbb{R})$ is called a *one-parameter unipotent subgroup* if U is the image of $\mathbb{R}w$ under the exponential map, for some matrix $w \in \mathrm{Mat}_{nn}$ satisfying $w^n = 0$ (that is, w is nilpotent and $\exp(tw)$ has only 1 as an eigenvalue, hence the name). For example, there is an index two subgroup $H \leqslant \mathrm{SO}(2,1)$ which is generated by one-parameter unipotent subgroups. However, notice that the diagonal subgroup A is not generated by one-parameter unipotent subgroups.

Raghunathan conjectured that if the subgroup H is generated by one-parameter unipotent subgroups, then the closures of orbits xH are always of the form xL for some closed connected subgroup L of G that contains H. This reduces the properties of orbit closures (a dynamical problem) to the algebraic problem of deciding for which closed connected subgroups L the orbit xL is closed.

Ratner [305] proved this important result using methods from ergodic theory. In fact, she deduced Raghunathan's conjecture from Dani's conjecture(10) regarding H-invariant measures, which she proved first in the series of papers [302, 303] and [304].

To date there have been numerous applications of the above theorem, and certain extensions of it. To name a few more seemingly unrelated applications, Elkies and McMullen [82] have applied these theorems to obtain the distribution of the gaps in the sequence of fractional parts of $\sqrt{n}$, and Vatsal [367] has studied values of certain L-functions using the p-adic version of the theorems. There are further applications of the theory too numerous to describe here, but the examples above show again the variety of fields that have connections to ergodic theory.

We will discuss a few special cases of the conjectures of Raghunathan and Dani. Example 1.3, Sect. 4.4, Chap. 10, Sect. 11.5, and Sect. 11.7 treat special cases, some of which were known before the conjectures were formulated.

1.8 An Overview of Ergodic Theory

Having seen some statements that qualify as being ergodic in nature, and some of the many important applications of ergodic theory to number theory, in this short section we give a brief overview of ergodic theory. If this is not already clear, notice that it is a rather diffuse subject with ill-defined boundaries(11).

Ergodic theory is the study of long-term behavior in dynamical systems from a statistical point of view. Its origins therefore are intimately connected with the time evolution of systems modeled by measure-preserving actions of the reals or the integers, with the action representing the passage of time. Related approaches, using probabilistic methods to study the evolution of systems, also arose in statistical physics, where other natural symmetries—typically reflected by the presence of a $\mathbb{Z}^d$-action—arise. The rich interaction between arithmetic and geometry present in measure-preserving actions of (lattices in) Lie groups quickly emerged, and it is now natural to view ergodic theory as the study of measure-preserving group actions, containing but not limited to several special branches:

(1) The classical study of single measure-preserving transformations.
(2) Measure-preserving actions of $\mathbb{Z}^d$; more generally of countable amenable groups.
(3) Measure-preserving actions of $\mathbb{R}^d$ and more general amenable groups, called flows.
(4) Measure-preserving and more general actions of groups, in particular of Lie groups and of lattices in Lie groups.

Some of the illuminating results in ergodic theory come from the existence of (counter-)examples. Nonetheless, there are many substantial theorems. In addition to fundamental results (the pointwise and mean ergodic theorems themselves, for example) and structural results (the isomorphism theorem of Ornstein, Krieger's theorem on the existence of generators, the isomorphism invariance of entropy), ergodic theory and its way of thinking have made dramatic contributions to many other fields.

Notes to Chap. 1

(1)(Page 1) The origins of the word 'ergodic' are not entirely clear. Boltzmann coined the word *monode* (unique μὸνος, nature εΐδος) for a set of probability distributions on the phase space that are invariant under the time evolution of a Hamiltonian system, and *ergode* for a monode given by uniform distribution on a surface of constant energy. Ehrenfest and Ehrenfest (in an influential encyclopedia article of 1912, translated as [78]) called a system *ergodic* if each surface of constant energy comprised a single time orbit—a notion called *isodic* by Boltzmann (same ισος, path ὁδός) — and *quasi-ergodic* if each surface has dense orbits. The Ehrenfests themselves suggested that the etymology of the word *ergodic* lies in a different direction (work έργον, path ὁδός). This work stimulated interest in the mathematical foundations of statistical mechanics, leading eventually to Birkhoff's formulation of the *ergodic hypothesis* and the notion of systems for which almost every orbit in the sense of measure spends a proportion of time in a given set in the phase space in proportion to the measure of the set.

(2)(Page 2) Questions of this sort were raised by Gel'fand; he considered the vector of first digits of the numbers $(2^n, 3^n, 4^n, 5^n, 6^n, 7^n, 8^n, 9^n)$ and asked if (for example) there is a value of $n > 1$ for which this vector is $(2, 3, 4, 5, 6, 7, 8, 9)$. This circle of problems is related to the classical Poncelet's porism, as explained in an article by King [194]. The influence of Poncelet's book [292] is discussed by Gray [126, Chap. 27].

(3)(Page 5) See also the account with some simplifications by Goldston, Motohashi, Pintz, and Yıldırım [117] and the survey by Goldston, Pintz and Yıldırım [119].

(4)(Page 8) In a more general form, this is the 11th of Hilbert's famous set of problems formulated for the 1900 International Congress of Mathematics.

(5)(Page 8) Bachet conjectured the result, and Diophantus stated it; there are suggestions that Fermat may have known it. The first published proof is that of Lagrange in 1770; a standard proof may be found in [87, Sect. 2.3.1] for example.

(6)(Page 9) For *indefinite* quadratic forms there is a very successful algebraic technique, namely strong approximation for algebraic groups (an account may be found in the monograph [286] of Platonov and Rapinchuk), so ergodic theory does not enter into the discussion.

(7)(Page 9) Under an additional congruence condition on Q' the method also works for $m \leqslant n - 3$.

(8)For some of the statements made here one actually has to assume that Γ is a *lattice*; see Sect. 9.4.3.

(9)(Page 9) Further readings from various perspectives on the ergodic theory of homogeneous spaces may be found in the books of Bekka and Mayer [21], Feres [90], Starkov [350], Witte Morris [385, 387] and Zimmer [394].

(10)(Page 10) For linear groups over local fields, and products of such groups, the conjectures of Dani (resp. Raghunathan) have been proved by Margulis and Tomanov [253] and independently by Ratner [306].

(11)(Page 11) Some of the many areas of ergodic theory that we do not treat in a substantial way, and other general sources on ergodic theory, may be found in the following books: the connection with information theory in the work of Billingsley [31] and Shields [342]; a wide-ranging overview of ergodic theory in that of Cornfeld, Fomin and Sinaĭ [60]; ergodic theory developed in the language of joinings in the work of Glasner [116]; more on the theory of entropy and generators in books by Parry [277, 279]; a thorough development of the fundamentals of the measurable theory, including the isomorphism and generator theory, in the book of Rudolph [324].